MASJID SOUND SOLUTIONS

The Masjid Sound Guide

Five reasons a prayer hall is hard to hear in — and what actually fixes each one.

Masjid Sound Solutions — a division of AZ Audio Solutions

Almost every masjid audio complaint comes back to one of five causes. None of them are solved by turning the volume up. This guide explains what is physically happening in the room, and what a correct fix looks like, so you can judge any proposal you are given — including ours.

01 The room is the loudest thing in it

Prayer halls are built from the hardest materials we have: tile, marble, plaster, glass, and often a dome. Sound does not stop at those surfaces, it bounces. What reaches the back rows is the imam plus several hundred milliseconds of the imam arriving late, layered on top of himself. That smearing is what people describe as "echo" or "I can hear him but I cannot understand him."

> **WHAT IS ACTUALLY HAPPENING**
>
> Reverberation time (RT60) is how long sound takes to decay by 60 dB. Speech needs a short RT60; a large hard-surfaced hall can easily run past two seconds, which is fine for a recitation and poor for a khutba. A dome makes it worse in a specific way: a curved ceiling focuses reflections back down into a hot spot instead of scattering them.

WHAT A CORRECT FIX LOOKS LIKE

- Measure RT60 before buying any equipment. If the room is the problem, no speaker upgrade will fix it.
- Treat the first-reflection surfaces — the rear wall behind the congregation and the side walls near the front — not the whole room.
- Carpet is not acoustic treatment. It absorbs the high frequencies you need for consonants and leaves the low-mid boom behind, which can make intelligibility worse.
- Under a dome, treat or diffuse the focal area rather than aiming more power at it.

02 One loud pair of speakers cannot cover a hall

The most common installation is two speakers at the front, turned up until the back row can just about hear. The front rows are then uncomfortably loud, the middle is acceptable, and the back is still straining. Turning it up further makes the front worse without meaningfully helping the back.

WHAT IS ACTUALLY HAPPENING

Sound pressure falls roughly 6 dB every time you double the distance from the source. Across a long hall that is a 15–20 dB swing between the first and last row. You cannot close that gap with power; you close it by adding sources so nobody is ever far from one.

WHAT A CORRECT FIX LOOKS LIKE

- Distribute more speakers at lower individual levels rather than fewer at high level. The target is even coverage, commonly specified as within ±3 dB across the seating area.
- Time-align distant speakers with DSP delay. Sound travels roughly one foot per millisecond, so an untreated rear speaker arrives ahead of the front sound and produces a distinct doubling on speech.
- Aim speakers at the congregation, not down the length of the room. Energy that hits a bare wall becomes the problem in section 01.
- Ask for a coverage plan — a drawing showing predicted level at seated ear height. If nobody can produce one, the layout is a guess.

03 Feedback is a gain-structure problem

The squeal when the imam steps toward a speaker, or when the wireless handheld is carried into the hall, is usually treated by turning the microphone down — which quietly removes the headroom the system needed in the first place.

WHAT IS ACTUALLY HAPPENING

Feedback happens when the microphone hears its own amplified output loudly enough to sustain a loop. The two variables that matter are how close the microphone is to the person speaking, and how much speaker energy lands back on the microphone. Both are placement decisions made before any processing is applied.

WHAT A CORRECT FIX LOOKS LIKE

- Get the microphone close to the source. Halving the mic-to-mouth distance is worth about 6 dB of usable gain, which is more than any processor will give you.
- Use a directional pattern (cardioid or hypercardioid) and orient the null toward the nearest speaker.

- **Never place a speaker behind or beside the microphone position. A speaker in front of the mic line is the single biggest structural fix.**
- Correct feedback with narrow parametric or notch filters at the specific ringing frequencies. Broad tone-control cuts sacrifice speech clarity to buy a little headroom.
- Automatic feedback suppressors are a safety net for a correctly designed system, not a substitute for one.

04 Speech intelligibility lives in the mid-band

Systems are often tuned by ear to sound impressive — bass lifted, top end lifted — and the result is a warm, expensive-sounding system that people still cannot follow a khutba on.

> **WHAT IS ACTUALLY HAPPENING**
>
> Consonants are what carry meaning, and they sit roughly between 1 kHz and 4 kHz. Vowels carry the volume. Boosting the low end adds loudness and energy that competes with the consonants, especially in a reverberant room where low frequencies decay slowest.

WHAT A CORRECT FIX LOOKS LIKE

- High-pass every speech microphone, typically around 100–120 Hz. Nothing below that is the voice; it is handling noise, HVAC rumble, and foot traffic.
- Tune speech and music as separate presets. The curve that flatters a nasheed will bury a khutba.
- Judge a tuning by whether unfamiliar words are intelligible from the back row, not by whether it sounds full from the mixer position.
- Where the budget allows, ask for a measured intelligibility figure (STI or %ALcons) rather than a subjective sign-off.

05 A system that volunteers cannot break

A well-tuned system drifts within months. Someone adjusts a knob for a wedding, someone else compensates the next day, and by Ramadan the settings bear no relation to the tuning that was paid for.

> **WHAT IS ACTUALLY HAPPENING**
>
> A masjid is not one venue. Jummah at capacity, daily salah with thirty people, weekend classes, an outdoor Eid prayer, and a livestream feed are five different requirements sharing one set of speakers. Without recallable presets, every one of those becomes a manual adjustment by whoever is available.

WHAT A CORRECT FIX LOOKS LIKE

- Insist on zones: main hall, sisters area, overflow and lobby, outdoor, and a separate feed for recording or livestream, each with its own level.

- **Set up recallable presets for the modes you actually run, selectable from a simple wall panel or phone app.**
- Lock the installer-level settings behind a password so the tuning cannot be edited by accident.
- Keep the DSP configuration file and the passwords in the masjid office, not only with the installer.

What to ask before you sign anything

- [ ] Will you measure the room before proposing equipment, and show me the measurements?
- [ ] Can I see a coverage plan showing predicted level across the seating area?
- [ ] Which surfaces are you treating acoustically, and why those?
- [ ] How many zones, and what presets will be programmed?
- [ ] Will you re-measure after installation and give me the results?
- [ ] Who holds the DSP file and the admin password when the job is finished?
- [ ] What is covered by warranty, for how long, and what is the response time for a Friday failure?

Every room is different, and none of the above replaces measuring yours. If you would like us to look at your space, or simply to sanity-check a proposal you have already received, we are glad to help either way.

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